

The h-likelihood

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HGLM course at the Roslin Institute

Definition of the h-likelihood

Lee & Nelder's (1996) hierarchical log-likelihood (**h-likelihood**):

$$h(\beta, \theta, u) = \log f(y|u) + \log f(u)$$

fixed effects β , variance components θ and random effects u .

The h-likelihood for a linear mixed model

For a linear mixed model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \mathbf{e}$ with $\mathbf{u} \sim N(0, \mathbf{I}_k \sigma_u^2)$ and $\mathbf{e} \sim N(0, \mathbf{I}_n \sigma_e^2)$

All we need is the normal density function (n iid observations with mean 0):

$$\log(f(\mathbf{y})) = \log \left[\frac{1}{(\sqrt{2\pi}\sigma^2)^n} \cdot e^{-\frac{1}{2\sigma^2}\mathbf{y}'\mathbf{y}} \right] = C - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \mathbf{y}'\mathbf{y}$$

We have $\mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}$ and

$$\begin{aligned} h(\boldsymbol{\beta}, \theta, \mathbf{u}) &= \log(f(\mathbf{e})) + \log(f(\mathbf{u})) \\ &= \left\{ C_1 - \frac{n}{2} \log(\sigma_e^2) - \frac{1}{2\sigma_e^2} \mathbf{e}'\mathbf{e} \right\} + \left\{ C_2 - \frac{k}{2} \log(\sigma_u^2) - \frac{1}{2\sigma_u^2} \mathbf{u}'\mathbf{u} \right\} \end{aligned}$$

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h-likelihood estimation for HGLM

Estimating fixed and random effects: $\frac{\partial h}{\partial \beta} = 0$ and $\frac{\partial h}{\partial u} = 0$

Estimating variance components using adjusted profile likelihood:

$$h_p = \left(h + \frac{1}{2} \log |2\pi \mathbf{D}^{-1}| \right)_{\beta = \hat{\beta}, u = \hat{u}}$$

$$\frac{\partial h_p}{\partial \theta} = 0$$

where $\mathbf{D}$ is the matrix of second derivatives of h around $\beta = \hat{\beta}, u = \hat{u}$. θ is a vector including both variance components, $\theta = (\sigma_u^2, \sigma_e^2)$.

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Deriving the hglm algorithm from the h-likelihood

For the linear mixed model we have

$$h = C - \frac{n}{2} \log(\sigma_e^2) - \frac{1}{2\sigma_e^2} \mathbf{e}'\mathbf{e} - \frac{k}{2} \log(\sigma_u^2) - \frac{1}{2\sigma_u^2} \mathbf{u}'\mathbf{u}$$

where $\mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}$.

Putting the first derivatives equal to zero (ie $\frac{\partial h}{\partial \boldsymbol{\beta}} = 0$ and $\frac{\partial h}{\partial \mathbf{u}} = 0$) gives Henderson's Mixed Model Equations.

Estimating the variance components: $\frac{\partial h_p}{\partial \theta} = 0$

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Deriving the hglm algorithm from the h-likelihood (cont.)

When we take the first derivative of $\log|\mathbf{D}|$, the hat values for the augmented model appears!

$$\frac{\delta}{\delta \sigma_e^2} \log|\mathbf{D}| = \text{tr}(\mathbf{D}^{-1} \frac{\delta}{\delta \sigma_e^2} \mathbf{D}) = \frac{-1}{(\sigma_e^2)^2} \text{tr}([\mathbf{X}, \mathbf{Z}]' [\mathbf{X}, \mathbf{Z}] (\mathbf{T}' \mathbf{T})^{-1}) =$$

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where we can recall that $\mathbf{T}$ is the design matrix for the augmented linear model. So, for the residual variance we have

$$\frac{\partial h_p}{\partial \sigma_e^2} = -\frac{n}{2\sigma_e^2} + \frac{1}{2(\sigma_e^2)^2} \sum_{i=1}^n \hat{e}_i^2 - \frac{1}{(\sigma_e^2)^2} \sum_{i=1}^n h_{ii}$$

This can be re-written as the score function for a gamma GLM with $\frac{\hat{e}_i^2}{1-h_{ii}}$ as response.

And similarly for $\sigma_u^2 \dots$

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Definition of the h-likelihood

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$$h(\beta, \theta, u) = \log f(y|u) + \log f(u).$$

- Classical inference
 - Uses marginal likelihood (random effects integrated out) $\int f(y, u) du$
 - Includes fixed parameters and only observations are treated as random
- Bayesian inference
 - Probabilistic framework that combines likelihood and prior information.
 - Treats all parameters and observations as random.
- Extended likelihood inference
 - The h-likelihood is based on the “extended likelihood principle”
 - Assuming simple statistical principles Bjørnstad (1996) showed that all information in the data about the random and fixed effects is included in a joint likelihood (which the h-likelihood is an implementation of)
 - The book by Pawitan (2001) gives an excellent introduction to the extended likelihood principle.
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Classical likelihood inference

A solution to inference for fixed unknowns θ was proposed by Fisher (1922). He developed a likelihood theory, expressing the probability to observe the data as the function of the parameter value.

Consider a statistical model, consisting of two types of objects, data $\mathbf{y}$ and parameter θ , and two related processes on them:

- **Statistical Model for Data Generation:** Generate an instance of the data $\mathbf{y}$ from a probability function with fixed parameters θ , $f_{\theta}(\mathbf{y})$.
- **Statistical Inference:** Given the data $\mathbf{y}$, make an inference about unknown fixed θ in the stochastic model by using the likelihood $L(\theta; \mathbf{y})$.
- The connection between these two processes is:

$$L(\theta; \mathbf{y}) \equiv f_{\theta}(\mathbf{y})$$

where L and f are algebraically identical, but on the left-hand side $\mathbf{y}$ is fixed while θ varies and on the right-hand side θ is fixed while $\mathbf{y}$ varies.

However, this approach avoids inference of any random effects.

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Classical marginal likelihood for a linear mixed model

Consider the linear mixed model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \mathbf{e}$$

In the classical likelihood approach the random effects are integrated out

$$f_{\theta}(\mathbf{y}) = \int f_{\theta}(\mathbf{u})f_{\theta}(\mathbf{y}|\mathbf{u})d\mathbf{u}$$

and the data generation process is given by a multivariate normal distribution $N(\mathbf{X}\boldsymbol{\beta}, \sigma_u^2\mathbf{Z}\mathbf{Z}' + \sigma_e^2\mathbf{I})$, with likelihood

$$L(\theta; \mathbf{y}) = (2\pi|\mathbf{V}|)^{-0.5} \exp(-0.5(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}))$$

Note: random effect $\mathbf{u}$ not included.

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Inference

For model comparisons and testing Lee & Nelder (1996) proposed to use

- the h-likelihood for random effects, $f_{\theta}(y, v)$
- the marginal likelihood for fixed effects, $f_{\theta}(y)$
- for the dispersion parameters $f_{\theta}(y|\hat{\beta})$

When the Laplace approximation is used, $f_{\theta}(y)$ is replaced by $p_v(h)$ and $f_{\theta}(y|\hat{\beta})$ by $p_{\beta, v}(h)$.

- **Model selection for nested HGLMs:** We can use the above likelihoods to perform likelihood ratio tests
- **Model selection for non-nested HGLMs:** The conditional AIC (cAIC) is defined as $-2f(y|v) + p_D$ where p_D are the estimated number of parameters (computed from the trace of the hat matrix).

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hglm package example

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data(semiconductor)
m14 <- hglm(fixed = y ~ x1 + x3 + x5 + x6,
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            family = Gamma(link = log),
            disp = ~ x2 + x3, data = semiconductor,
            calc.like = TRUE)

summary(m14)
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h-likelihood:   -614.3952
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EQL estimation converged in 4 iterations.

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Extended likelihood

- Birnbaum (1962) proved that the classical likelihood function contains all the information about the value of the fixed parameter.
- Bjornstad (1996) extended this concept and showed that all information in the data $\mathbf{y}$ for parameters θ and unobservables $\mathbf{u}$ is in the extended likelihood.

Extended likelihood

- **Statistical Model for Data Generation:**

- (i) Generate an instance of the random quantities $\mathbf{u}$ from a probability function $f_{\theta}(\mathbf{u})$ and then with $\mathbf{u}$ fixed,
- (ii) generate an instance of the data $\mathbf{y}$ from a probability function $f_{\theta}(\mathbf{y}|\mathbf{u})$.
- The combined stochastic model is given by the product of the two probability functions $f_{\theta}(\mathbf{u})f_{\theta}(\mathbf{y}|\mathbf{u})$

- **Statistical Inference:** Given the data $\mathbf{y}$, we can

- (i) make inferences about θ by using the marginal likelihood $L(\theta; \mathbf{y}) \equiv f_{\theta}(\mathbf{y})$,
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h-likelihood

- Lee & Nelder (1996) proposed the use of the hierarchical (h-)likelihood, an extended likelihood defined on a pre-defined scale of $\mathbf{u}$, for inferences for both fixed and random unknowns.
- If there exists a method, which gives a better inference than that using the extended likelihood, then the method uses an information not in the data, provided the assumed model is true (Bjornstad 1996).
- Note that in the absence of fixed parameters θ , the extended likelihood becomes proportional to the Bayesian posterior, while in the absence of unobservables it becomes the classical likelihood.